

5th Progress Report

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Date:	22/02/2024
Subject:	RL for Flying-V v2.0
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Project term:	01/2024 - 08/2024
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Outcome of the previous meeting

- Reviewed IDHP extensions, should make additions to notation to make explicit definitions of variables (time-step when they're evaluated, critic & actor variables), need to change the critic error to be consistent with Jun and Stefan implementations.
- Reviewed research questions. Should change the 1st question to make 3rd & 4th sub-questions more research-oriented, and add a sub-question in the 2nd question to address nominal control challenges.
- Set target for delivering literature review + research plan sometime at end of 7th week or middle of 8th week (23rd to 28th of Feb).
- Got CitAst manual for compiling simulink models for use in Python.

Description of the results since last meeting

- Written half of the literature study and made rl taxonomy chart (see Figure 1), currently chapter is sectioned as follows:
 1. Basic RL theories (completed)
 2. DP & ADP (completed)
 3. DRL (in progress)
 4. Flying-V
 5. Reinforcement learning for flight control
 6. Conclusion
- Drafted research plan.
- Updated research questions (included in research plan).
- Updated Gantt chart (attached in research plan).

Plans for next 2 weeks

1. Review project plan in a future meeting.
2. Continue writing literature study.

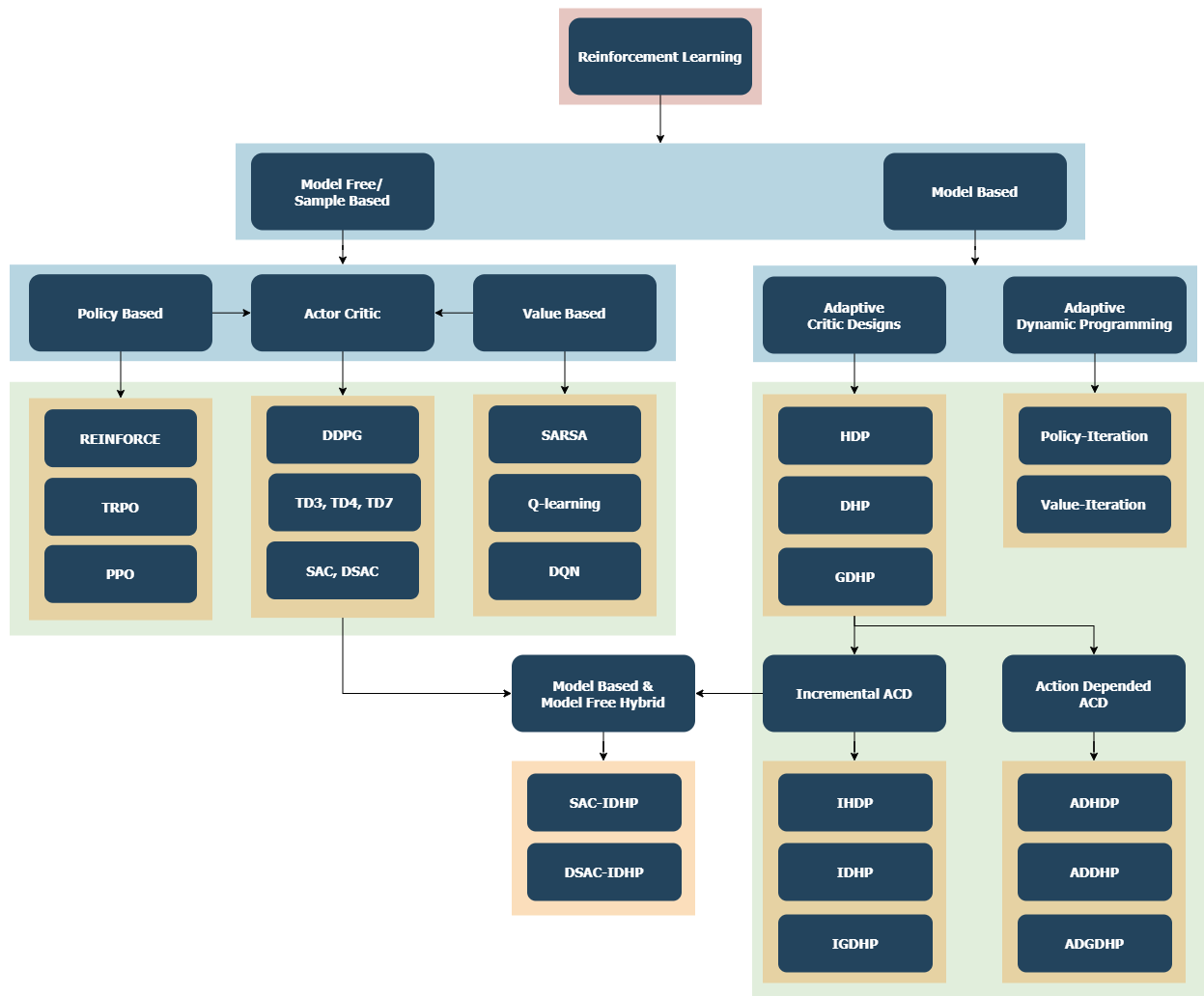


Figure 1: RL taxonomy, the chart was made before I made better charts in my literature study, and the style of the chart will be updated later to match the charts in the literature study